Acoustic features with forward and backward selection (k = 10)

FINAL_V1 experiment group summary

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Results

- 12 of 12 conditions in this group are complete.
- Highest pooled AUPRC (descriptive, selection-biased among 12 conditions): AC Logistic regression, AUPRC 1.000, AUROC 1.000.
- Lowest: AC/BACKWARD Logistic regression, AUPRC 0.990, AUROC 0.987.
- Range of AUROC over the group: 0.985 to 1.000; median 0.996.
- Speaker errors at threshold 0.5 (false negatives + false positives) per condition: minimum 1, median 1, maximum 4 (out of 47 speakers).
- Fold variability: median over conditions of the mean fold AUROC is 1.000 with a median SD of 0.000 over the five outer folds (9 to 10 speakers per fold).

Note on the selection table

- The selector is a fixed logistic regression (C = 1), so the selected features in a fold do not depend on the final model; the four rows of each method are therefore identical (verified in the audit).

Selection versus all 40 features (same model)

- AUROC difference (selected minus all), median -0.004, range -0.013 to 0.007; interval excludes zero in 0 of 8 comparisons (descriptive).

Reading the table

- Primary metrics are pooled speaker-level out-of-fold AUPRC and AUROC (47 speakers: 25 PD, 22 controls; prevalence 53.2% PD); F1, sensitivity, specificity and balanced accuracy use the fixed threshold 0.5.
- Intervals are percentile bootstrap intervals over speakers (2,000 resamples); with 47 speakers one person moves a metric by roughly 2 points. Comparisons are descriptive; no significance is claimed.
- Recording date and class are confounded in this corpus (every date holds one class; all 44.1 kHz audio is PD), so a high score does not show disease detection.

Table 1: Pooled out-of-fold speaker-level results

Representation	Model	AUPRC [95% CI]	AUROC [95% CI]	F1	Sens.	Spec.	Bal.acc.
AC	Logistic regression	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.980	0.960	1.000	0.980
AC	Linear SVM	0.998 [0.991, 1.000]	0.998 [0.989, 1.000]	0.980	0.960	1.000	0.980
AC	RBF SVM	0.991 [0.971, 1.000]	0.989 [0.964, 1.000]	0.923	0.960	0.864	0.912
AC	XGBoost	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.980	0.960	1.000	0.980
AC/FORWARD	Logistic regression	0.997 [0.985, 1.000]	0.996 [0.983, 1.000]	0.980	0.960	1.000	0.980
AC/FORWARD	Linear SVM	0.997 [0.985, 1.000]	0.996 [0.983, 1.000]	0.980	0.960	1.000	0.980
AC/FORWARD	RBF SVM	0.997 [0.987, 1.000]	0.996 [0.984, 1.000]	0.960	0.960	0.955	0.957
AC/FORWARD	XGBoost	0.997 [0.985, 1.000]	0.996 [0.983, 1.000]	0.980	0.960	1.000	0.980
AC/BACKWARD	Logistic regression	0.990 [0.966, 1.000]	0.987 [0.957, 1.000]	0.941	0.960	0.909	0.935
AC/BACKWARD	Linear SVM	0.994 [0.977, 1.000]	0.993 [0.971, 1.000]	0.980	0.960	1.000	0.980
AC/BACKWARD	RBF SVM	0.990 [0.963, 1.000]	0.985 [0.947, 1.000]	0.923	0.960	0.864	0.912
AC/BACKWARD	XGBoost	0.996 [0.982, 1.000]	0.995 [0.978, 1.000]	0.980	0.960	1.000	0.980

Table 2: Selection stability across the five outer folds (features chosen in all 5 folds / distinct features ever chosen; most frequent)

Method	Model	In 5/5	Distinct	Most frequent features
FORWARD	Logistic regression	0	18	log_energy_db_mean (4/5), log_energy_db_p10 (4/5), zcr_mean (4/5), zcr_p10 (4/5)
FORWARD	Linear SVM	0	18	log_energy_db_mean (4/5), log_energy_db_p10 (4/5), zcr_mean (4/5), zcr_p10 (4/5)
FORWARD	RBF SVM	0	18	log_energy_db_mean (4/5), log_energy_db_p10 (4/5), zcr_mean (4/5), zcr_p10 (4/5)
FORWARD	XGBoost	0	18	log_energy_db_mean (4/5), log_energy_db_p10 (4/5), zcr_mean (4/5), zcr_p10 (4/5)
BACKWARD	Logistic regression	3	18	band4_log_mean (5/5), band4_log_p10 (5/5), band4_log_p90 (5/5), band3_log_p10 (4/5)
BACKWARD	Linear SVM	3	18	band4_log_mean (5/5), band4_log_p10 (5/5), band4_log_p90 (5/5), band3_log_p10 (4/5)
BACKWARD	RBF SVM	3	18	band4_log_mean (5/5), band4_log_p10 (5/5), band4_log_p90 (5/5), band3_log_p10 (4/5)
BACKWARD	XGBoost	3	18	band4_log_mean (5/5), band4_log_p10 (5/5), band4_log_p90 (5/5), band3_log_p10 (4/5)